

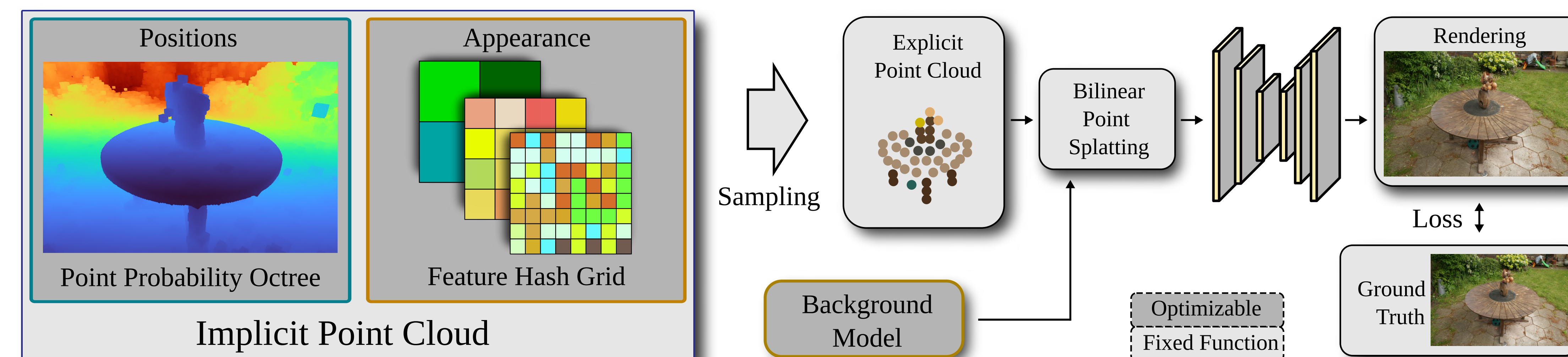
Abstract

We introduce a new approach for reconstruction and **novel view synthesis** of **unbounded real-world scenes**.

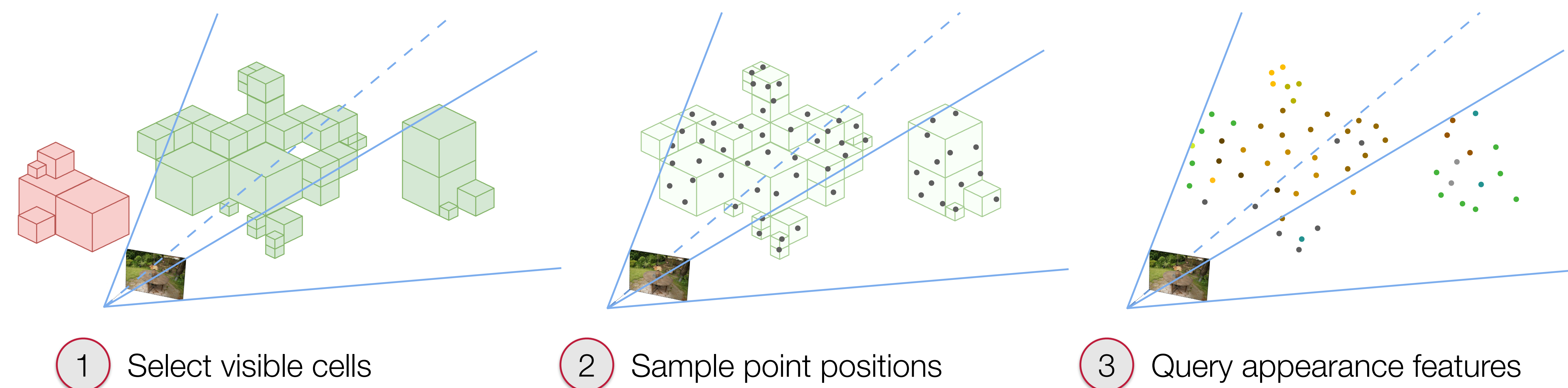
In contrast to previous methods using either volumetric fields, grid-based models, or discrete point cloud proxies, we propose a **hybrid scene representation**, which implicitly encodes the **geometry** in a continuous **octree-based probability field** and **view-dependent appearance** in a **multi-resolution hash grid**. This allows for extraction of arbitrary explicit point clouds, which can be rendered using rasterization. In doing so, we combine the benefits of both worlds and retain favorable behavior during optimization: Our novel **implicit point cloud** representation and differentiable bilinear rasterizer enable fast rendering while preserving the fine geometric detail captured by volumetric neural fields. Furthermore, this representation does not depend on priors like structure-from-motion point clouds.

Our method achieves **state-of-the-art image quality** on common benchmarks. Furthermore, we achieve fast inference at **interactive frame rates**, and can convert our trained model into a large, explicit point cloud to further enhance performance.

Rendering Pipeline



Point Cloud Sampling



Results

Method	Mip-NeRF360 [1]			Tanks and Temples [5]			Train hrs↓	Render fps↑
	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑		
TRIPS [3]	0.213	0.778	25.94	0.229	0.831	22.62	4.00	90.0
3DGS [4]	0.254	0.814	27.20	0.276	0.866	25.27	0.50	194.2
Zip-NeRF [2]	0.219	0.828	28.56	0.233	0.878	26.75	5.00	0.2
Ours	0.164	0.847	28.56	0.189	0.878	25.93	9.58	2.9
Ours (16M)	0.173	0.841	28.36	0.220	0.862	25.12	6.15	5.5
Ours (8M)	0.188	0.830	27.83	0.252	0.846	24.54	4.25	9.4
Ours (pre-ex.)	0.207	0.802	26.85	0.261	0.833	23.71	4.25	27.7



In our user study, participants preferred **INPC** over **Zip-NeRF** of **69.41%** of the time.

Method	Sampling	Rendering	Post-Proc.	Total	#Points
Ours (8M)	51 ms	37 ms	26 ms	114 ms	4×8M
Ours (8M)	16 ms	10 ms	26 ms	52 ms	8M
Ours (pre-ex.)	-	28 ms	26 ms	54 ms	67M

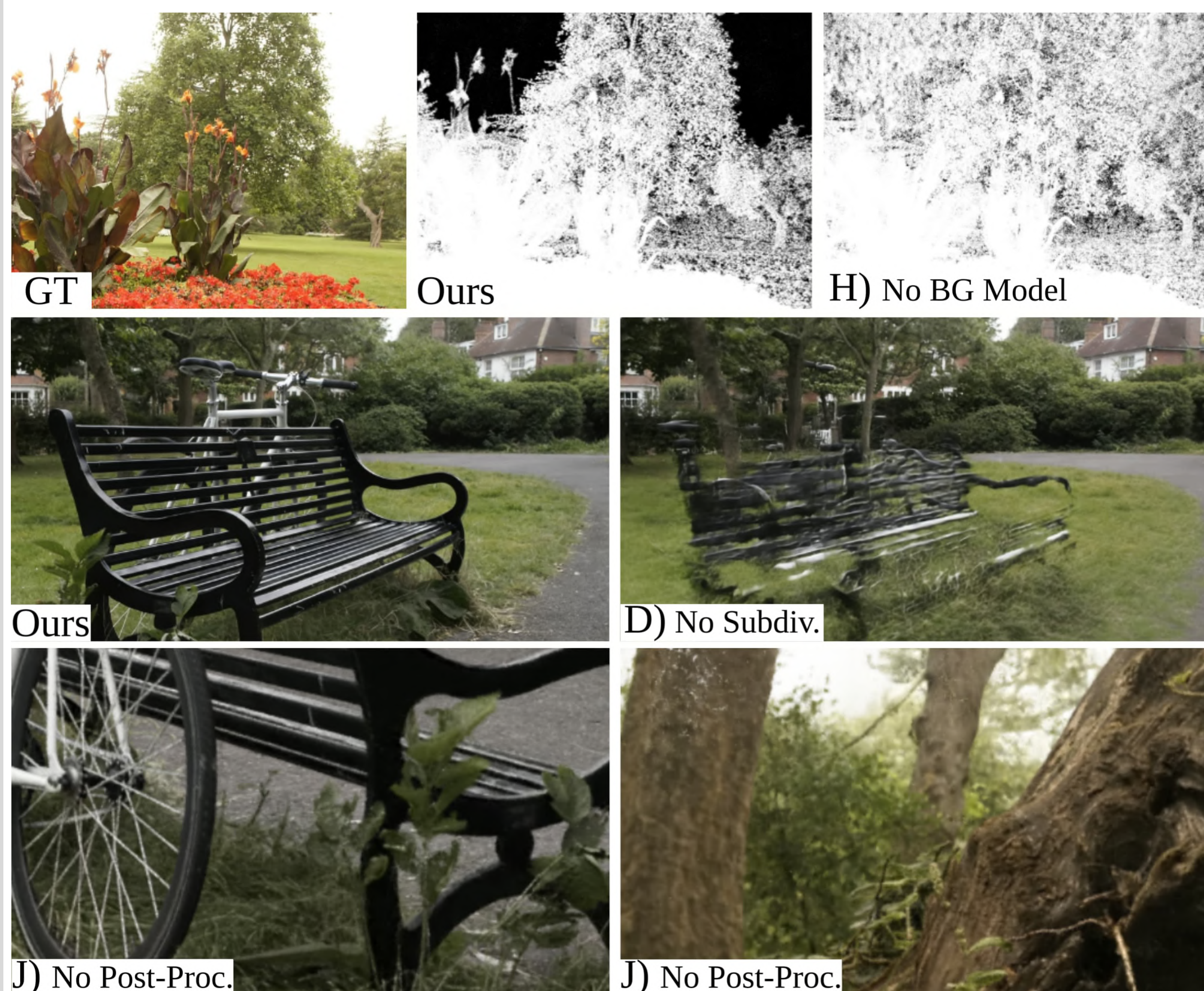
Resolution: 2000×1085 GPU: RTX 4090

The rendering bottleneck is sorting, which requires 27 / 7 / 17 ms respectively.



INPC beats **Zip-NeRF** in terms of image quality and renders 10× faster. Explicit point-based approaches are faster but achieve lower quality.

Ablation Study



Visual Comparisons



✓ Better reconstruction of fine detail

✓ Less prone to under-reconstruction

References

- [1] Barron et al. Mip-NeRF 360, CVPR 2022.
- [2] Barron et al. Zip-NeRF, ICCV 2023.
- [3] Franke et al. TRIPS, CGF 2024.
- [4] Kerbl et al. 3D Gaussian Splatting, TOG 2023.
- [5] Knapitsch et al. Tanks and Temples, TOG 2017.

Funding



Project Page

